In an IPv4 network, one of the routers have 4 links numbered 0 through

Destination address range	Outgoing interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
default 3	

How does the router forward datagrams that arrive with the following destination IP addresses?
Show your work.
(a) 192.145.81.85
(b) 225.64.195.60
(c) 193.128.17.119

There are two TCP senders. The first sender uses TCP *Tahoe*, the second uses *Reno*. Assume that the MSS is 1 KB, that the one-way propagation delay for both connections is 50 ms and that the link joining the two routers has a bandwidth of 6 Mb/s. Let *cwnd*1 and *cwnd*2 be the values of the senders' congestion windows.

a) What is the smallest value of *cwnd*1+ *cwnd*2 for which the link joining the two routers stays busy all the time?
The RTT is 100 ms in this case, so the link rate is equivalent to 600 Kb per RTT or 75 KB. So, cwnd1+cwnd2= 75 KB.

b) Assume that the link buffer overflows whenever *cwnd*1+*cwnd*2≥150 KB and that at time 0, *cwnd*1=30 KB and *cwnd*2=120 KB. Approximately, what are the values of *cwnd*1 and *cwnd*2 one RTT later? Also, what are the values of *ssthresh* for each of the two connections? Assume that all losses are detected by triple duplicate ACKs.

Since the first uses Tahoe and the second uses Reno, cwnd1=1 KB and cwnd2= 60 KB, ssthresh1=15KB and sshresh2= 60 KB.

c) After 8 more RTTs, approximately what are the values of *cwnd*1 and *cwnd*2?
19 KB and 68 KB

d) Approximately, how many more RTTs before *cwnd*1+*cwnd*2≥150 KB again?
(150–87)/2=31.5 RTTs

e) What is *cwnd*2–*cwnd*1 at this point?
*cwnd*2–*cwnd*1*=49 KB*

Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 90; the second has sequence number 110.

a. How much data is in the first segment?
b. Suppose that the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, what will be the acknowledgment number?

Suppose that the five measured SampleRTT values are 106 ms, 120 ms, 140 ms, 90 ms, and 115 ms. Compute the EstimatedRTT after each of these SampleRTT values is obtained, using a value of $\alpha = 0.125$ and assuming that the value of EstimatedRTT was 100 ms just before the first of these five samples were obtained. Compute also the DevRTT after each sample is obtained, assuming a value of $\beta = 0.25$ and assuming the value of DevRTT was 5 ms just before the first of these five samples was obtained. Last, compute the TCP TimeoutInterval after each of these samples is obtained.